Model vs Index — CAGR Comparison

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What period this covers

Item	Value
Test window	2023-07-01 to 2025-01-31
Rebalance frequency	monthly
Rebalance dates	18
Elapsed time	1.50 years
Holding period	entry to the next rebalance's entry (chaining)
Portfolio	equal-weighted top 20, rebalanced each period
CAGR formula	product of (1 + period return) over 18 periods, then raised to 1/1.50

Basis

Price return compared with price return. NSE's historical index endpoint serves price indices only — every total-return (TRI) spelling returns no data. The strategy's dividends are therefore reported separately rather than folded into the headline, because comparing a dividend-inclusive strategy against a dividend-less index would credit the model with roughly the market's yield (about 1–1.5% a year in India) for nothing.

Index returns are measured over the sessions the strategy actually traded — the same next-session entry and the same exit — not between month-ends, so neither side is measuring a different span of market time.

Headline: CAGR by strategy

Strategy	Gross	Net	vs UNIVERSE	vs N50	vs N500	vs NMIDCAP 150	vs NSMALLCAP 250
equal weight universe	29.17%	27.66%	-1.51%	+15.75%	+11.30%	+3.49%	+3.83%
growth only	49.61%	46.97%	+17.80%	+35.06%	+30.61%	+22.80%	+23.14%
model 5	53.63%	46.37%	+17.20%	+34.46%	+30.01%	+22.20%	+22.54%
model 5 gated	39.79%	31.64%	+2.47%	+19.74%	+15.28%	+7.48%	+7.82%
momentum only	66.45%	52.99%	+23.82%	+41.08%	+36.63%	+28.82%	+29.16%
quality only	26.65%	25.47%	-3.70%	+13.56%	+9.11%	+1.31%	+1.64%
random ranking	28.91%	6.63%	-22.54%	-5.28%	-9.73%	-17.53%	-17.20%
simple quality momentum	31.99%	25.02%	-4.15%	+13.11%	+8.65%	+0.85%	+1.19%
value only	65.91%	60.25%	+31.07%	+48.34%	+43.88%	+36.08%	+36.42%

vs UNIVERSE is the column that matters, and it is deliberately placed before the index columns. It compares the strategy against an equal-weighted holding of the very same point-in-time universe it selected from, so it isolates stock selection.

Why the index columns flatter every strategy. The eligible universe returned 29.17% against NIFTY 500's 16.36% — a +12.81 point gap earned purely by equal-weighting a broad, small-cap-heavy universe, before any selection at all. That gap is inside every "vs index" number below. A strategy that beats NIFTY 500 by 20 points while matching the universe has selected nothing.

Index reference

Index	CAGR	Volatility	Sharpe	Max drawdown	Periods
NIFTY 50	11.91%	11.31%	1.05	-9.44%	18
NIFTY 500	16.36%	12.75%	1.26	-11.82%	18
NIFTY MIDCAP 150	24.17%	15.68%	1.47	-12.51%	18

NIFTY SMALLCAP 250	23.83%	18.76%	1.24	-15.40%	18
Eligible universe (equal weight)	29.17%	-	-	-	18

The **eligible universe** row is the benchmark that controls for size and breadth: the same point-in-time universe the strategy chose from, held equally weighted. Beating NIFTY 500 while trailing this means the model captured a small-cap tilt rather than stock selection.

Risk-adjusted comparison

Strategy	Basis	CAGR	Volatility	Sharpe	Max DD	Hit rate
equal weight universe	net	27.66%	20.09%	1.32	-12.75%	67%
growth only	net	46.97%	21.73%	1.90	-10.29%	83%
model 5	net	46.37%	22.45%	1.83	-12.32%	72%
model 5 gated	net	31.64%	18.38%	1.60	-11.41%	61%
momentum only	net	52.99%	27.62%	1.69	-11.19%	67%
quality only	net	25.47%	15.34%	1.56	-8.44%	67%
random ranking	net	6.63%	22.07%	0.39	-21.82%	39%
simple quality momentum	net	25.02%	19.03%	1.28	-13.25%	67%
value only	net	60.25%	28.45%	1.82	-19.71%	67%

Excess return against NIFTY 500

Strategy	Mean excess/period	Tracking error	Information ratio	Periods beaten
equal weight universe	+0.88%	15.05%	0.70	61%
growth only	+2.11%	20.64%	1.23	61%
model 5	+2.09%	21.71%	1.16	61%
model 5 gated	+1.12%	17.64%	0.76	61%
momentum only	+2.56%	23.95%	1.28	56%
quality only	+0.67%	12.34%	0.65	67%
random ranking	-0.61%	16.02%	-0.46	39%
simple quality momentum	+0.69%	15.50%	0.53	61%
value only	+2.98%	24.07%	1.49	61%

Information ratio is excess return per unit of tracking error — how reliably the outperformance was earned, not how large it was. **Periods beaten** is the share of rebalances where the strategy outpaced the index; a high CAGR with a low share was carried by a handful of periods.

Turnover and cost drag

Strategy	One-way turnover	Gross CAGR	Net CAGR	Cost drag
equal weight universe	0.066	29.17%	27.66%	1.51%
growth only	0.108	49.61%	46.97%	2.64%
model 5	0.295	53.63%	46.37%	7.26%
model 5 gated	0.410	39.79%	31.64%	8.14%
momentum only	0.492	66.45%	52.99%	13.46%
quality only	0.063	26.65%	25.47%	1.18%
random ranking	0.963	28.91%	6.63%	22.28%
simple quality momentum	0.368	31.99%	25.02%	6.97%
value only	0.163	65.91%	60.25%	5.66%

Cost drag is annualised, and it is the difference between a ranking that works on paper and one that works in an account. A strategy replacing most of its book every month pays it repeatedly.

Before quoting these numbers

An index is investable and this portfolio is a simulation. The index CAGR is achievable through a low-cost fund; the strategy CAGR assumes every fill happened at the modelled price and size.

The window is what it is. A CAGR over a few years is dominated by the market regime inside it. Check the periods-beaten share and the drawdown before treating a CAGR gap as skill.

Costs are modelled, not measured. Statutory charges are exact and effective-dated; half-spread and market impact are assumptions, which is why gross and net are both shown.

Delisting recovery is an assumption. Re-run with --delisting-strategy zero and last_close to see how much of the gap depends on it.